

Language Models (LMs)

Natural Language Processing

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Language modeling

- Language modeling refers to different probabilistic and statistical methods used to determine the chances of a given series of words occurring in a text. These models in language modeling analyze text and provide predictions as to which words might come next.
- <https://www.xavor.com/blog/language-modeling-in-nlp-a-comprehensive-overview/>

How it works

1. Training:

- Language models are trained on vast datasets like Wikipedia, books, or news articles. The model learns to understand the structure and patterns of the language from this data.

2. Prediction Process:

- When a model is given a sequence of words, it looks at the past words (context) and predicts the next word by selecting the one with the highest probability.

3. Evaluation:

- The performance of language models is often measured using a metric called **perplexity**, which indicates how well the model predicts the next word. Lower perplexity means better performance.

Language Models

- Models that assign probabilities to sequences of words
- Assigns a probability to each possible next word (or sentence).
- Predicts the next word given a context
 - “he briefed reporters on the main contents of the <next word?>”
- Which of the following is more likely?
 1. “all of a sudden I notice three guys standing on the sidewalk”
 2. “on guys all I of notice sidewalk three a sudden standing the”

Corpus statistics

i	want	to	eat	chinese	food	lunch	spend
2533	927	2417	746	158	1093	341	278

Example

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Figure 3.1 Bigram counts for eight of the words (out of $V = 1446$) in the Berkeley Restaurant Project corpus of 9332 sentences. Zero counts are in gray.

Example (probabilities)

	i	want	to	eat	chinese	food	lunch	spend
i	0.002	0.33	0	0.0036	0	0	0	0.00079
want	0.0022	0	0.66	0.0011	0.0065	0.0065	0.0054	0.0011
to	0.00083	0	0.0017	0.28	0.00083	0	0.0025	0.087
eat	0	0	0.0027	0	0.021	0.0027	0.056	0
chinese	0.0063	0	0	0	0	0.52	0.0063	0
food	0.014	0	0.014	0	0.00092	0.0037	0	0
lunch	0.0059	0	0	0	0	0.0029	0	0
spend	0.0036	0	0.0036	0	0	0	0	0

Figure 3.2 Bigram probabilities for eight words in the Berkeley Restaurant Project corpus of 9332 sentences. Zero probabilities are in gray.